

CÉDRIC ACCARD

Astrophysics PhD Student

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EDUCATION

2024 - 2025	First year of PhD in Astrophysics	Observatoire Astronomique de Strasbourg
2023 - 2024	Second year of Master's Degree in Astrophysics With high honours	Université de Strasbourg
2022 - 2023	First year of Master's Degree in Physics With high honours	Université de Strasbourg
2019 - 2022	Bachelor's Degree in Fundamental and Applied Physics (3 years) With high honours	Université de Lorraine
2019	High school diploma in Sciences With high honours	Lycée Jean Zay, Jarny

RESEARCH EXPERIENCE

02/24 - 06/24	Spatially-resolved Far-UV to Far-IR exploration of massive main-sequence galaxies at $z \sim 4.5$	Université de Strasbourg
- Context : 2nd year of master's degree internship at the Observatoire Astronomique de Strasbourg - Work : Using ALMA/HST/JWST observations of three $z \sim 4.5$ galaxies spanning wavelengths from UV to FIR, we aim to produce SED curves in order to retrieve physical properties of the objects and study the Kennicutt-Schmidt relation of those objects. Codes in Python / Used CIGALE code for SED modeling / Used CASA for data reduction		
01/23 - 04/23	Distribution of cold gas in galaxies one billion years after the Big Bang	Université de Strasbourg
- Context : 1st year of master's degree internship at the Observatoire Astronomique de Strasbourg - Work : Using high-resolution [CII] and continuum maps from ALMA as well as convolved Hubble ancillary data of four massive main-sequence galaxies at $z \sim 4.5$, we produced (un-)obscured star formation rate and gas surface density maps. Then using averaging method, we studied the Kennicutt-Schmidt relation for those objects to put constraints at such redshift. This has led to 2 publications in peer-review journals. Codes in Python / Used CASA for data reduction		
05/22	Possibilities for laser control of vibrations of a cold molecule: The case of RbSr	Université de Lorraine
- Context : 3rd year of bachelor's degree internship at the Institut de Chimie Physique et Matériaux Metz - Work : After a bibliographical research on laser control methods of diatomic molecule and it's theoretical background, I wrote a code to simulate the transition between states of the RbSr molecules via different methods. Also tested previous experimental results and gave interpretation for future works. Codes in Octave		

PEER-REVIEWED PUBLICATIONS

First Author	Cosmic evolution of the [CII] -to-molecular gas relation	C. Accard et al. 2026, Astronomy & Astrophysics, submitted - arXiv:2604.19529
Co-author	Evaluating star formation rates at $z=5$	D. Ismail et al., incl. C. Accard 2026, Astronomy & Astrophysics, accepted 01/04 - arXiv:2601.05916
Co-author	Starbursts hiding in the main sequence: a pathway toward quenching?	F. Renaud et al., incl. C. Accard 2026, Open Journal of Astrophysics, submitted - arXiv:2602.23445

Co-author	Multi-tracer exploration of molecular gas in main sequence galaxies at $z \sim 4.5$ M. Dessauges-Zavadsky et al., incl. C. Accard 2025, Astronomy & Astrophysics, in revision
First author	The ALPINE-CRISTAL-JWST survey: spatially resolved star formation relations at $z \sim 5$ C. Accard et al. 2025, Astronomy & Astrophysics, 702, A206
Co-author	Spatially resolved [CII]-gas conversion factor in early galaxies L. Vallini et al., incl. C. Accard 2025, Astronomy & Astrophysics, 700, A117
Co-author	The ALMA-ALPINE [CII] survey: Sub-kpc morphology of 3 main-sequence galaxy systems at $z \sim 4.5$ revealed by ALMA T. Devereaux et al., incl. C. Accard 2024, Astronomy & Astrophysics, 686, A156
Second author	The ALMA-ALPINE [CII] survey: Kennicutt-Schmidt relation in four massive main-sequence galaxies at $z \sim 4.5$ M. Béthermin, C. Accard, et al. 2023, Astronomy & Astrophysics, 680, L8

SKILLS

Languages:	French - <i>native</i> , English - <i>C1 level (TOEIC) and Master's degree done in english</i>
Coding :	Python, Octave, LaTeX